MIL-G-44108A
10 June 1987
SUPERSEDING
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30 June 1983

MILITARY SPECIFICATION

GLOVES, COMBAT VEHICLE CREWMAN'S, SUMMER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of lightweight, unlined, glove, fabricated from perspiration and water resistant leather and simplex knitted, fire resistant, polyamide cloth.

1.2 Classification. The gloves shall be furnished in the following sizes, as specified (see 6.2):

| SIZES | | | | | | |
|---|---|---|---|---|---|---|
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| A-A-50083 | - Bag, Plastic, Folded Garment |
| JJ-W-155 | - Webbing, Textile, (Cotton, Elastic) |
| KK-L-2004 | - Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned |
| DDD-L-20 | - Label: for Clothing, Equipage, and Tentage, (General Use) |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-L-35078 | - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For |
| MIL-T-43636 | - Thread, Aramid |
| MIL-C-81393 | - Cloth, Knitted, Polyamide, High Temperature Resistant, Simplex, Jersey |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-191 | - Textile Test Methods |
| FED-STD-311 | - Leather, Methods of Sampling and Testing |
| FED-STD-751 | - Stitches, Seams, and Stitchings |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized Unit Loads |
| MIL-STD-731 | - Quality of Wood Members for Containers and Pallets |

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)



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2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.4).

3.2.1 Leather. The leather shall be full grain cattlehide 2 1/2 to 3 ounces thick or full grain horsehide 1 3/4 to 2 1/2 ounces thick. The color of the leather shall be black. The cattlehide shall conform to type V and the horsehide to type VII of KK-L-2004, with the following exceptions: the leather shall be treated with glutaraldehyde to resist perspiration and treated with an approved water resistant compound (see 6.2) to resist water absorption, the colorfastness (resistance to crocking) shall be not lower than



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Munsell value 7.0 for staining of the dry cloth and 6.5 for staining of the wet cloth, and the minimum stitch tearing strength shall be 20.0 pounds. In addition to the applicable requirements of KK-L-2004, the leather shall meet the requirements specified in 3.2.1.1 through 3.2.1.7.

* 3.2.1.1 Staining. The stain on the test pad shall be not lower than expanded Chromatic Transference Scale value +2 when tested for staining in accordance with 4.4.1.1.

3.2.1.2 Area stability to perspiration. At least 80 percent of the specimens tested shall have an area loss no greater than 15 percent when tested as specified in 4.4.1.1.

3.2.1.3 Area stability to laundering. At least 80 percent of the specimens tested shall have an area loss no greater than 20 percent when tested as specified in 4.4.1.1.

3.2.1.4 Water absorption resistance. At least 80 percent of the specimens tested shall gain not more than 20 percent of their weight by water absorption when tested as specified in 4.4.1.1.

3.2.1.5 Oil absorption resistance. The leather shall show no evidence of oil absorption when tested as specified in 4.4.1.1.

3.2.1.6 Air permeability. Three hundred cubic centimeters of air shall pass through the leather in ten minutes or less when tested as specified in 4.4.1.1.

3.2.1.7 Thickness of cut parts. When cattlehide is used, the leather parts shall be 2 1/2 to 3 ounces thick. When horsehide is used, the leather parts shall be 1 3/4 to 2 1/2 ounces thick.

3.2.2 Cloth. The cloth shall conform to MIL-C-81393. The color of the cloth shall be Olive Green 106.

3.2.3 Elastic webbing. The elastic webbing shall conform to type II, class 1 (5/16 inch) or class 2 (3/8 inch), unbleached, bleached, or dyed (color optional) of JJ-W-155.

3.2.4 Thread. All the sewing operations shall be performed with aramid thread conforming to size B of MIL-T-43636. The color of the thread shall approximately match the color of the cloth.



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3.3 Design and construction. The back of the glove and thumb, including the base of the thumb and a portion of the gauntlet, and the four finger seamless fourchette shall be fabricated from the high temperature resistant, polyamide, simplex knitted cloth for protection from fire. The palm and the front of the gauntlet, thumb, and fingers of the glove shall be fabricated from one piece of the leather. The gauntlet of the glove shall contain an elastic webbing takeup. The gloves shall be constructed in accordance with figures 1 through 5. No cut edges, except where specified for the leather, shall be visible on the exterior of the glove. The grain side of the leather and the face side of the fabric shall be on the outside of the glove.

* 3.3.1 Patterns. The component parts of the glove shall be cut in strict accordance with the patterns furnished by the Government. The Government patterns shall not be altered in any manner and shall be used for making the working patterns or dies. The working patterns or dies shall be identical to the Government furnished patterns. Leather components shall be cut using standard commercial methods for pattern or table cut gloves. Clicker cut methods shall not be used for cutting the leather components. When applicable, blocker patterns shall be used for blocking out the tranks.

3.3.1.1 Pattern parts. The pattern parts of the gloves shall be as specified in table I.

TABLE I. Glove pattern parts

| Pattern nomenclature | Material | Parts per pair of gloves |
|---|---|---|
| Back of thumb | Simplex cloth | 2 |
| Back of hand and fingers | Simplex cloth | 2 |
| Continuous fourchette | Simplex cloth | 2 |
| Front of hand and thumb | Leather | 2 |

3.3.2 Cutting. The leather shall be examined on both sides prior to cutting, and any area that contains surface defects shall not be included in the cut leather parts. The leather shall then be sorted for weight and shade. The leather shall be dampened and the stretch pulled out in the length direction. Hard parts of the leather that do not stretch adequately shall not



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be cut into gloves. The tranks shall be cut so the stretch is in the width direction of the fabricated glove. The length (cuff to fingertip) direction of the cloth parts shall be cut in the wale direction of the fabric.

* 3.3.3 Seams and stitching. All seam construction shall be in accordance with FED-STD-751 and as shown on figures 1 through 4. No trimmer shall be used. The closing of the shape tabs of the index and little fingers, closing of the side seam, and attaching of the cloth fourchettes to the cloth back of the fingers shall be done using stitch type 401 or 301 and seam type SSa-1. The leather fingers shall be joined to the cloth fourchettes from finger tip of little finger to finger tip of the index finger using stitch type 401 and seam type LSa-1. The leather index finger shall be attached to the back of the glove from the tip of the finger to the top of the gauntlet using stitch type 401 or 301 and seam type LSa-1. The leather front of thumb and leather continuation of gauntlet shall be attached to the back of thumb with two rows of stitching 1/16 to 1/8 inch apart using stitch type 401 and seam type LSa-2 (see figures 1, 2, 3, and 4). For all LSa type seams, the flesh side of the leather shall be lapped over the face side of the fabric. The looper thread of the stitch type 401 stitching shall be on the innerside of the glove. The top of the glove shall be hemmed with stitch type 301. The elastic webbing shall be attached with zigzag stitching, stitch type 304. The beginning and ending of the zigzag row of stitching shall be tacked with 2 to 4 rows of vertical stitching (see figure 3). All the stitch types shall conform to FED-STD-751 and shall have 9 to 12 stitches per inch. Each row of stitching shall be straight and parallel to the seam edge. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. No seam or component shall be twisted, puckered, or pleated and no part of the glove shall be caught in an unrelated operation or seam. Unless otherwise specified, all the thread ends shall be trimmed to a length of not more than 1/8 inch. Loose thread ends shall be removed from the gloves. Ends of 301 stitching shall be backstitched or overstitched 1/2 inch minimum except where turned under in a hem or held down by other stitching. Both ends of all seams or stitching produced with 401 stitch type when not turned under in a hem or not held down by other stitching shall have a 1/4 to 3/4 inch chain extending beyond each end.

3.3.3.1 Repairs of types 301 and 401 stitching. Repairs of stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching.



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b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or tight stitching shall be repaired by removing the defective stitching without damaging the material and then restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.3.4 Attaching of the elastic webbing. The elastic webbing shall be centered and attached across the seam, on the inside of the glove, as in figures 1 and 3, section C-C. The webbing shall be stretched to its full length so that the shirrings on the outerside, where the webbing is attached, shall be evenly spaced and the full amount of the stretch in the webbing shall be utilized.

3.3.5 Forming. The thumb shall be well worked out on the heated thumb stick. The gloves shall be laid-off on heated forms of proper size. The glove shall be pulled down so that the side seams of all the fingers shall be straight and parallel to the side seam of the glove. The finger tips shall be fully extended to remove all wrinkles, creases, or lumps which may affect the comfort. The forming shall not stiffen or scorch the cloth or leather.

3.3.6 Pairing. The gloves shall be matched and paired. The paired gloves shall be tacked together with thread, 1/2 to 1 inch from the top of the glove, in the cloth portion of the back of the thumb and gauntlet continuation. The tacking shall be accomplished with not less than two stitches, using a ply thread, which shall permit the gloves to be easily pulled apart, without tearing or damaging the cloth.

3.4 Dimensions. The finished gloves shall conform to the dimensions indicated in table II. All dimensions shall be determined in accordance with 4.4.3. The variation in the length and width, between the left and right glove of any pair, shall not exceed 1/4 inch.



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TABLE II. Glove dimensions in inches

| Location | Size 5 | 6 | 7 | 8 | 9 | 10 | 11 | Tolerance |
|---|---|---|---|---|---|---|---|---|
| A | 12 1/4 | 12 1/2 | 12 3/4 | 12 7/8 | 13 1/8 | 13 3/8 | 13 1/2 | ± 3/16 |
| B | 2 3/4 | 2 13/16 | 3 | 3 1/16 | 3 1/8 | 3 3/16 | 3 1/4 | ± 1/8 |
| C | 3 7/16 | 3 1/2 | 3 9/16 | 3 5/8 | 3 11/16 | 3 3/4 | 3 13/16 | ± 1/8 |
| D | 2 13/16 | 2 7/8 | 3 1/16 | 3 1/8 | 3 3/16 | 3 1/4 | 3 5/16 | ± 1/8 |
| E | 2 1/4 | 2 5/16 | 2 7/16 | 2 /2 | 2 9/16 | 2 5/8 | 2 11/16 | ± 1/8 |
| F | 2 5/16 | 2 13/16 | 3 | 3 1/8 | 3 1/4 | 3 1/4 | 3 3/8 | ± 1/8 |
| G | 3 5/8 | 3 3/4 | 3 7/8 | 4 | 4 1/8 | 4 1/4 | 4 3/8 | ± 3/16 |
| Stretch | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | Minimum |

3.5 Marking.

3.5.1 Identification. The cloth portion of each glove shall be marked in accordance with type III, class 4 of DDD-L-20 except that the color of the identification markings shall be white or any contrasting color. The markings shall withstand fastness to laundering as specified in DDD-L-20. The markings shall be located on the inside of each glove, on the back of the gauntlet, approximately 1 inch from the hem. The heat, used to transfer the markings, shall not stiffen or scorch the cloth or leather. The markings shall be as follows:

| | |
|---|---|
| GLOVES, COMBAT VEHICLE CREWMAN'S, SUMMER | S |
| Contract No. | I |
| NSN | Z |
| (Contractor's Name) | E |

THE CLOTH PORTION OF THE GLOVE IS FABRICATED FROM AN INHERENTLY FIRE RESISTANT MATERIAL (NOMEX) THAT DOES NOT MELT OR DRIP. CAN BE LAUNDERED WITHOUT LOSING ITS FIRE RESISTANT PROPERTIES AND NO RETREATMENT IS NECESSARY.



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3.5.2 Instruction. A lightweight, white, paper, instruction label with legible and durable black characters shall be inserted into each right hand glove in such a manner as to prevent loss when being handled or packaged. The size of the paper label shall be approximately 2 1/2 inches by 5 inches. The label shall contain the following information.

WASHING INSTRUCTIONS - GLOVES, COMBAT VEHICLE CREWMAN'S, SUMMER

1. The gloves may be washed with a mild soap in water not over 120°F in temperature.

2. Rinse thoroughly, then stretch into shape.

3. Drip dry or wrap in a towel. Do not wring or twist dry. Do not expose to hot air or full sunlight.

3.6 Replacement of defective components. During the spreading, cutting, and manufacturing processes, components having material defects or damages that are classified as defects in 4.4.2 and 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

3.7 Workmanship. After completion of the final assembly, the gloves shall be thoroughly cleaned and all thread scraps, lint, and foreign matter shall be removed. The gloves shall not contain any spot, stain, cut, abraded area, tear, mend, needle chew, or ragged edge. The gloves shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The gloves shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component testing. The leather shall be tested in accordance with table III for the characteristics shown. The methods of testing specified in FED-STD-311, and FED-STD-191 wherever applicable, and as listed in table III shall be followed. The leather sampling procedure and location from which the sample unit is to be obtained shall be in accordance with the sampling for testing of non-fabricated leather as defined in the sampling for inspection



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section of FED-STD-311, except that the test area for obtaining the test specimens shall be 12 inches by 12 inches for each unit. The lot shall be rejected if any one of the following conditions exist:

a. Any specimen fails to meet the requirements for staining.
b. More than three specimens fail to meet the requirements for area stability to perspiration or area stability to laundering or water absorption.
c. Any specimen fails to meet the requirements for oil absorption resistance or air permeability.
d. Treatment certification is missing.

TABLE III. Component tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Treatment | 3.2.1 | 1/ |
| Staining | 3.2.1.1 | 3021 |
| Area stability to perspiration | 3.2.1.2 | 3211 |
| Area stability to laundering | 3.2.1.3 | 3041 2/ |
| Water absorption resistance | 3.2.1.4 | 4.5.1 |
| Oil absorption resistance | 3.2.1.5 | 4.5.2 |
| Air permeability | 3.2.1.6 | 5452 3/ |

1/ The contractor shall certify that the leather has been treated with glutaraldehyde and that an approved water resistant compound was used to treat the leather.



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2/ Except that one specimen shall be tested from each sample unit.

3/ Refers to FED-STD-191. The leather specimen shall be mounted with the grain side toward the inner cylinder so the air will pass through the specimen from grain side to flesh side. One specimen shall be tested from each sample unit and results shall be reported as pass or fail.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. Defects for pairing shall be classified as a single defect.

NOTE: When an asterisk appears in the classification column, the defect shall be classified as major when it seriously affects the serviceability or appearance and minor when it does not seriously affect either serviceability or appearance.

TABLE IV. End item visual defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| General | Any hole (except needle hole), cut, tear, mend, or burn | X | |
| | Any weakening defect such as a smash, multiple float, loose slub, needle chew, or abraded area | * | |
| | Needle holes | | |
| | - More than ten | X | |
| | - Five to ten | | X |
| | Color of any component not as specified, not uniform, any outerside part shaded, or contains any shade bar | * | |
| | Two or more thread ends not trimmed or two or more thread scraps not removed from the glove | | X |
| | Any spot or stain | * | |



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TABLE IV. End item visual defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| General - continued | Any portion of the leather or cloth stiffened or scorched by any process of manufacturing | X | |
| | Two or more ragged edges on the outer side of the glove | | X |
| Leather | Grain missing | * | |
| | Wrinkles or loose, spongy, or boardy leather | * | |
| | Not soft and pliable | * | |
| | Grain cracked, peeled, or abraded | * | |
| | Scar, scratch, brittle area, bony area, or thin spot | * | |
| | Any other grain damage or surface imperfection | * | |
| Knit cloth | Any run, dropped stitch, snag, pull, or slubby yarn 1/ | * | |
| | Lacking elasticity, too tight or loose | * | |
| | Wales not in length direction | X | |
| Components and assembly | Any component not as specified or any defect of assembly | * | |
| | Any component, component part, or required operation omitted, or any operation improperly performed | * | |
| | Grain surface of the leather or face of the cloth not on the outer surface of the glove | X | |
| Seams and stitching | Accuracy of seaming: | | |
| | - Any seam, hem, or attachment of any component puckered, pleated, or caught in an unrelated operation or stitching | X | |
| | - Any seam allowance not as specified | * | |



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TABLE IV. End item visual defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Seams and stitching - continued | Gage of stitching: 2/ | | |
| | - Any stitching irregular, unevenly gaged or not gaged as specified, or not uniform | * | |
| | - Distance between the two rows of stitching attaching the thumb and gauntlet continuation of the thumb less than 1/32 inch or more than 5/32 inch | X | |
| | Any open seam 3/ | X | |
| | Seam and stitch type: | | |
| | - Not specified seam or stitch | * | |
| | - Any looper thread on the outer side of any glove seam, when stitch type 401 is used | X | |
| | Stitch tension: | | |
| | - Any loose stitching 4/ | * | |
| | - Any tight stitching resulting in breaking of the stitches when normal pull is applied 5/ | X | |
| | Stitches per inch: 6/ | | |
| | - Stitches more or less than specified: | | |
| | - More than two stitches | X | |
| | - Two stitches | | X |
| Pairing | Not right and left glove of the same size, or wide variation in appearance | X | |
| | Not tacked or joined together as specified | | X |
| Marking | Any missing, or size incorrect, incomplete, or illegible | X | |
| | Other than the size information, incorrect, incomplete, or illegible | | X |
| | Improperly located or any strike-through | * | |



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1/ One or two single pick runs on the innerside of a glove shall not be classified.

2/ The gage of stitching defect shall be classified only when the condition exists on the major portion of the seam.

3/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur. On double stitched seams, a seam shall be classified as open when either one or both sides of the seam contain any of the aforementioned defects for a single stitched seam.

4/ A seam shall be classified as loose when the top and bobbin threads are on the same surface.

5/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified as an accuracy of seaming defect.

6/ The stitches per inch defect shall be classified only when the condition exists on the major portion of the seam.

4.4.3 Dimensional examination of gloves. The gloves shall be examined for dimensional defects including glove stretch and thickness of leather thumb, fingers, palm, and gauntlet. Any dimension not meeting the specified requirement shall be classified as a defect. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Thickness of leather parts shall be determined in accordance with Method 1011 of FED-STD-311. Width and length measurements shall be made while the gloves are placed on a flat surface with wrinkles and creases smoothed out. No tension shall be applied when making width and length measurements. Measurements shall be made as specified in table V.



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TABLE V. Methods of measurement

| Location (see figure 5) | Method |
|---|---|
| A | Measure the overall length, from the tip of the second finger to the top of the glove |
| B | Measure the length of the index finger, from the base to the tip of the finger |
| C | Measure the length of the second finger, from between the base of the index and second fingers to the tip of the second finger |
| D | Measure the length of the third finger, from between the base of the second and third fingers to the tip of the third finger |
| E | Measure the length of the fourth (little) finger, from between the base of the third and fourth fingers to the tip of the fourth finger |
| F | Measure the length of the thumb, from the pipe of the thumb to the tip of the thumb |
| G | Measure the width, on the palm side, 1/4 inch below the base of the third and fourth fingers |

Stretch shall be determined by grasping the glove, with the thumb and the forefinger at the fold-line side and the closing seam side, at the location where the width dimension G was taken. While held in this position, the glove shall be subjected to an outward pull (stretch) to the maximum extent that can be exerted by hand pressure, without damaging the glove. The width of the glove after stretching shall be determined where width dimension G was taken. The difference between the original width dimension and the width dimension after stretching shall be recorded as the glove stretch.



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4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing, damaged, or not as specified |
| Workmanship | Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling<br>Open and noncontinuous heat-sealed seams of polyethylene bag<br>Incorrectly fabricated polyethylene bag<br>Bulged or distorted container |
| Content | Number of pairs of gloves per shipping container is more or less than required<br>Size shown on one or more pairs of gloves not as specified on shipping container 1/<br>Number of pairs of gloves per intermediate container is more or less than required 1/<br>Number of intermediate containers per container is more or less than required |

1/ For this defect, two intermediate containers shall be examined from each container in the sample.

* 4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.



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| Examine | Defect |
|---|---|
| Finished dimensions | Length, width, or height exceeds specified maximum requirement |
| Palletization | Pallet pattern not as specified<br>Interlocking of loads not as specified<br>Load not bonded as specified |
| Weight | Exceeds maximum load limits |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |

4.5 Methods of inspection.

4.5.1 Water absorption resistance. One 3-inch diameter specimen shall be tested from each 12 inch by 12 inch test area. Prior to testing, the specimen shall be allowed to reach moisture equilibrium in accordance with the atmospheric conditions for testing section of FED-STD-311. A stainless steel 500 mL, 3 inch diameter, 5 inch long container (see 6.5) with a rubber gasket and cover shall be used to test the specimen. The weighed specimen shall be placed inside the cover with the flesh side of the specimen next to the inside of the cover. Place the rubber gasket inside the cover and against the grain surface of the leather specimen. Add 100 mL of distilled water at 23°C to the container. Secure the cover with specimen and gasket to the container. Invert the container so the water is against the grain surface of the specimen. Place the inverted container on a flat surface and allow it to remain in the inverted position for 30 minutes. At the end of 30 minutes place the container upright and remove the cover and place it on a flat surface. The inside of the cover shall face the flat surface. Tilt the cover so the outer edge of the bottom is raised 1 to 1 1/2 inches from the flat surface and allow it to drain for 5 minutes. After 5 minutes, immediately remove the specimen from the cover, hold the specimen by its edge and shake briskly once to remove any free moisture from the surface of specimen then immediately weigh the specimen to the nearest 0.01 gram. Calculate the percent water absorption as follows:

$$\text{Percent water absorption} = \frac{(W_2 - W_1)}{W_1} \times 100$$

where: $W_1$ = Original weight of conditioned specimen, and

$W_2$ = Weight of specimen after being subjected to water.



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4.5.2 Oil absorption resistance. One 4 inch by 4 inch specimen shall be tested from each 12 inch by 12 inch test area. Place the specimen on a horizontal flat surface flesh side up. With a dropping bottle pipette, carefully place (do not drop) three small drops (approximately 3/16 inch diameter or 0.05 mL volume per drop) of Nujol mineral oil, or equal, on the flesh surface of the specimen. Each drop shall be in a different location. After 5 minutes immediately observe the drops from an angle of approximately 45° for evidence of penetration of the oil into the leather at the oil-leather interface or wicking of the oil into the leather at the edges of the drops. The mineral oil shall have a Saybolt viscosity of 360/390 at 100°F; specific gravity of 0.880/0.900 at 60°F (see 6.6).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Unit packing. Each pair of gloves shall be placed in a snug-fitting bag conforming to A-A-50083.

5.1.1.2 Intermediate packing. Six pairs of the bagged gloves of one size only, with every other pair in a reverse position to the one on top of it, shall be packed within a commercial type intermediate folding or set up box, approximately 13 1/2 inches by 5 1/2 inches by 2 inches. The boxes shall be sealed with kraft paper tape. The tape shall be applied lengthwise on the seams and shall extend beyond the seams for at least 2 inches. Full telescopic boxes, if used, need not be sealed.

5.1.2 Commercial. The gloves shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Sixteen boxes, 96 pairs of gloves of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Inside dimensions of each shipping container shall approximate 27 inches in length, 11 1/2 inches in width and 8 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there is less



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than the required amount per container of the same size, mixed sizes may be packed within the same shipping container. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Sixteen boxes, 96 pairs of gloves of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Inside dimensions of each shipping container shall approximate 27 inches in length, 11 1/2 inches in width and 8 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there is less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Gloves, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), gloves, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 35 in accordance with appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate packs, and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged gloves. Polyethylene bagged gloves shall have the required identification information legibly printed or stamped in yellow directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.



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5.4.2 Labels, mixed sizes. Each shipping container, packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 inches by 4 inches, with the words "MIXED NSN's" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity of pairs and NSN's contained therein.

6. NOTES

6.1 Intended use. The gloves are intended to be worn by combat vehicle crewmen in the warm temperate zones.

6.2 Ordering data. Acquisition documents shall specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. List of approved water resistant compounds (see 3.2.1).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.1).

6.5 Container. The type I, 500 mL stainless steel container used with the Atlas Launder-Ometer is suitable for testing the leather for water absorption. The container can be purchased from the Atlas Electric Devices Co., 4114 N. Ravenswood Avenue, Chicago, IL 60613. The manufacturer's part No. is 11-2432.

6.6 Mineral oil. Nujol is a mineral oil available in most drug stores and has been found to be suitable for use in testing the leather for oil absorption resistance. Nujol is the trademark of Plough, Inc.



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* 6.7 Subject term (key word) listing.

Clothing
Gloves, combat
Protective clothing

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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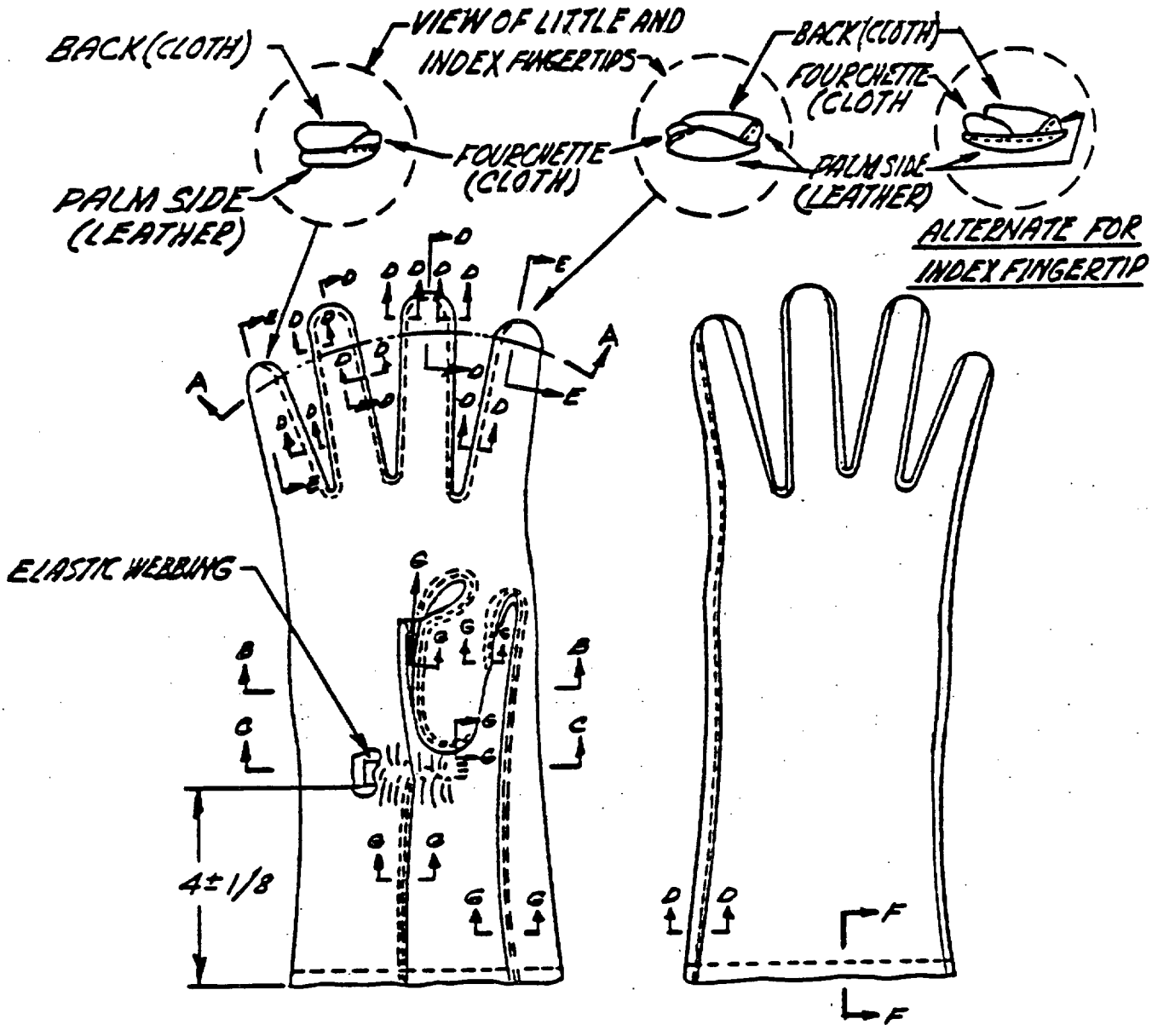
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED.
TOLERANCES ON DIMENSIONS SHALL BE AS FOLLOWS

| DIMENSIONS | TOLERANCES |
|---|---|
| TO 1/2 INCLUSIVE | +1/16 -0 |
| OVER 1/2 TO 1 INCLUSIVE | ±1/16 |
| OVER 1 TO 6 INCLUSIVE | ±1/8 |

FOR SECTION A-A, SEE FIGURE 2
FOR SECTIONS B-B AND C-C SEE FIGURE 3.
FOR SECTIONS D-D, E-E, F-F AND G-G SEE FIGURE 4.

Figure 1. Gloves, Combat Vehicle Crewman's, Summer.



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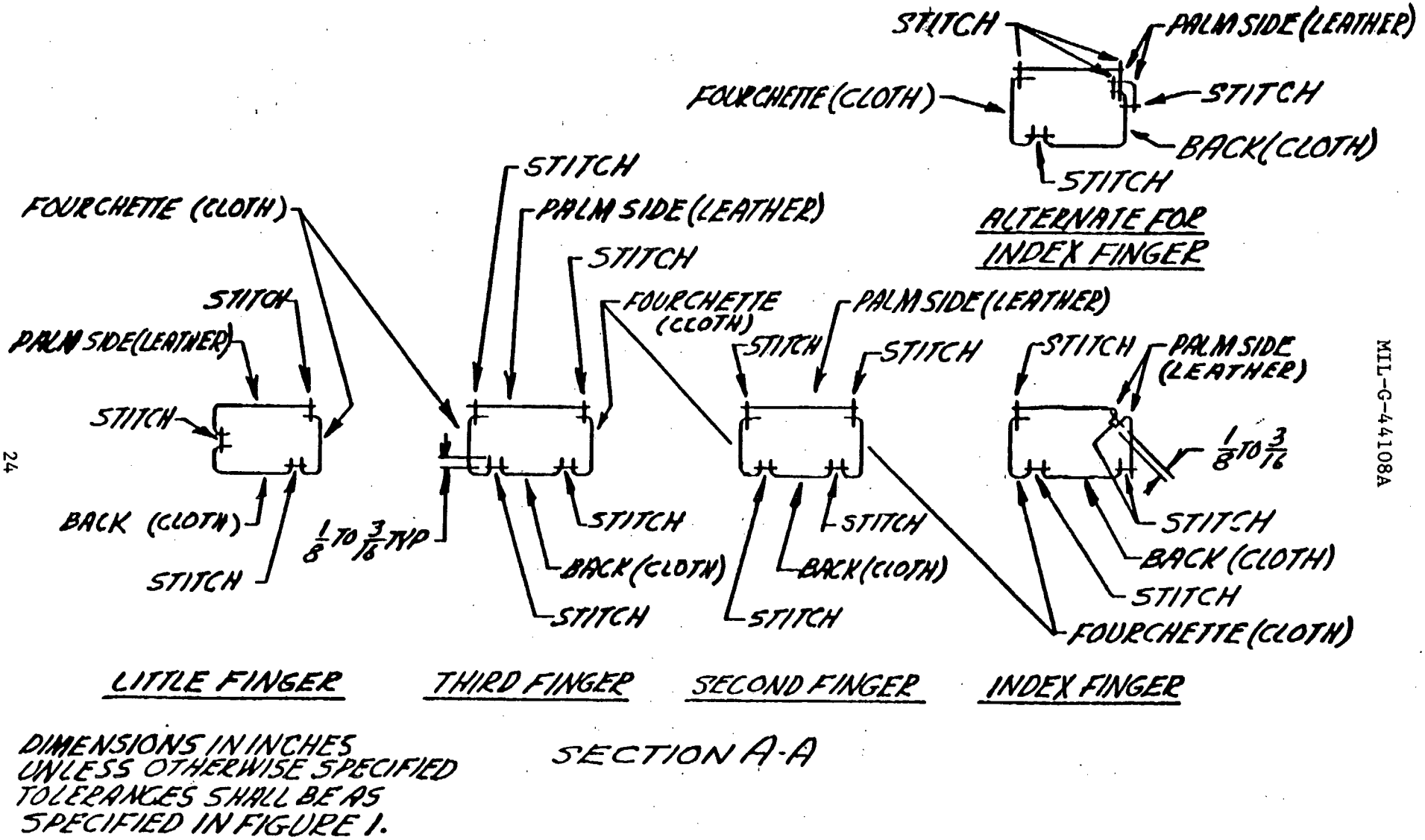



Figure 2. Gloves, Combat Vehicle Crewman's, Summer.



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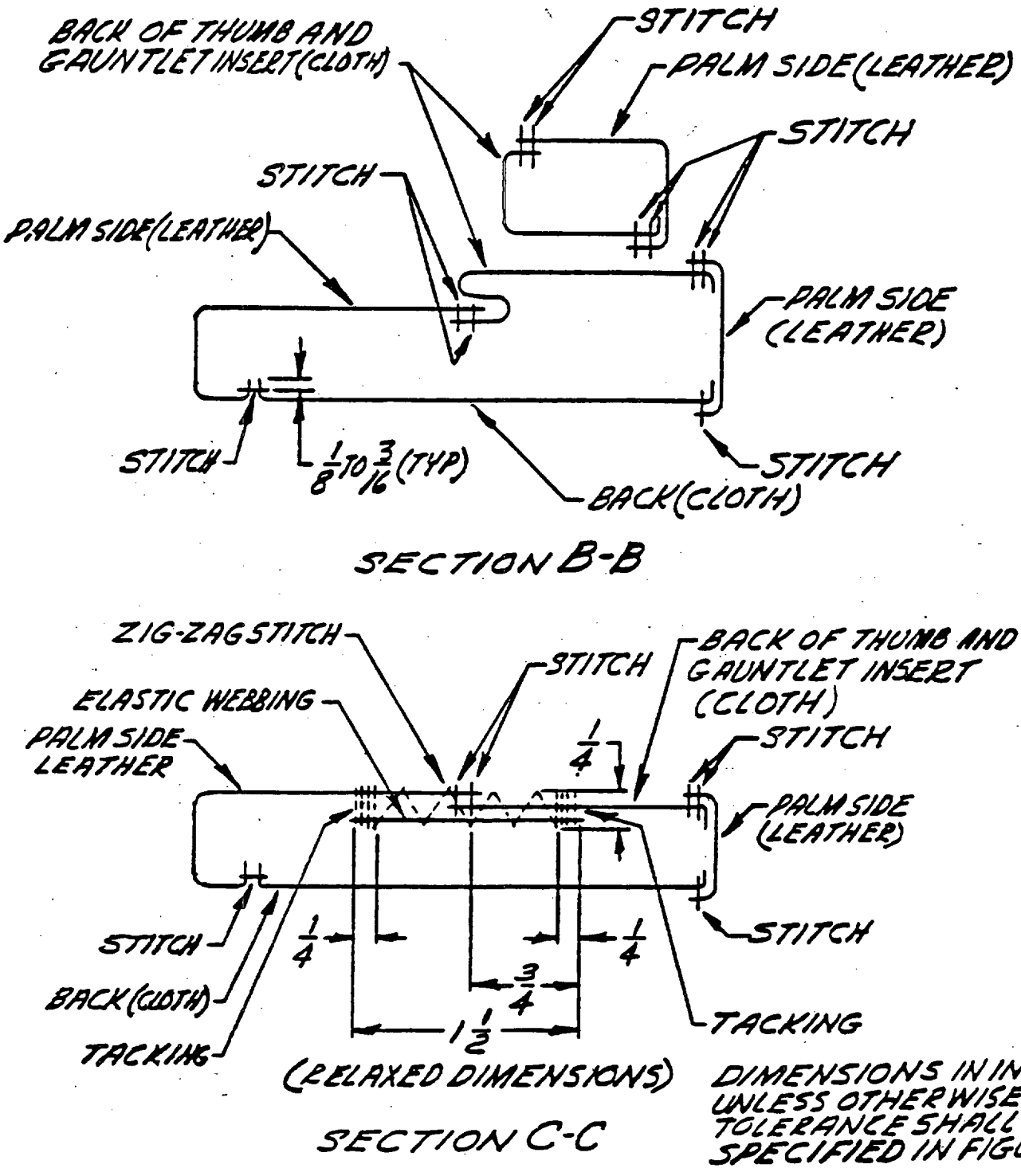
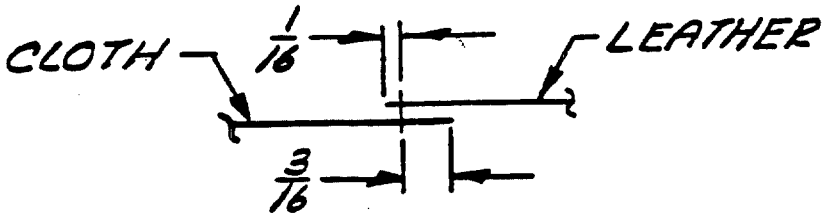


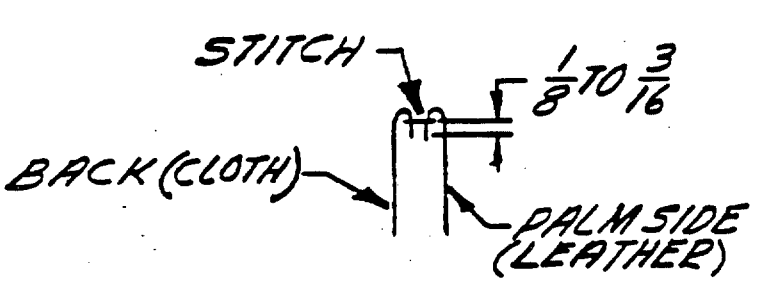

Figure 3. Gloves, Combat Vehicle Crewman's, Summer.



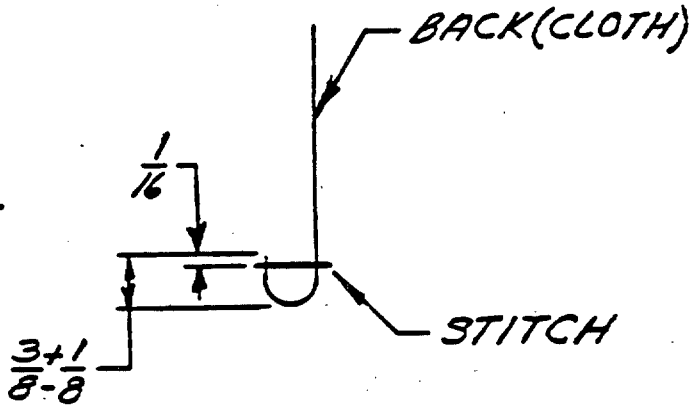
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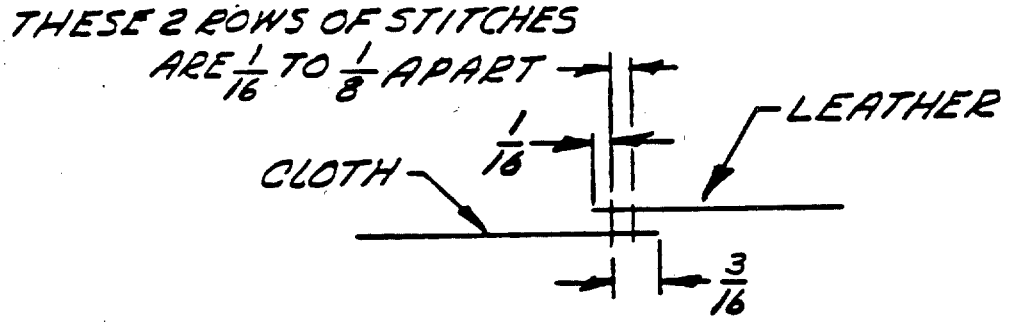

SECTION D-D
TYPICAL SEAM JOINING THE LEATHER TO THE FOURCHETTE (CLOTH)
SEE NOTE




SECTION E-E




SECTION F-F




SECTION G-G
TYPICAL SEAM JOINING THE THUMB AND GAUNTLET
CONTINUATION OF THE THUMB (SEE NOTE)

'MENSIONS IN INCHES.
VLESS OTHERWISE SPECIFIED,
'LERANCES SHALL BE AS
PECIFIED IN FIGURE 1.

NOTE: FOR REFERENCE, THE OVERLAP OF THE LEATHER SHALL BE 1/4 INCH.

Figure 4. Gloves, Combat Vehicle Crewman's, Summer.



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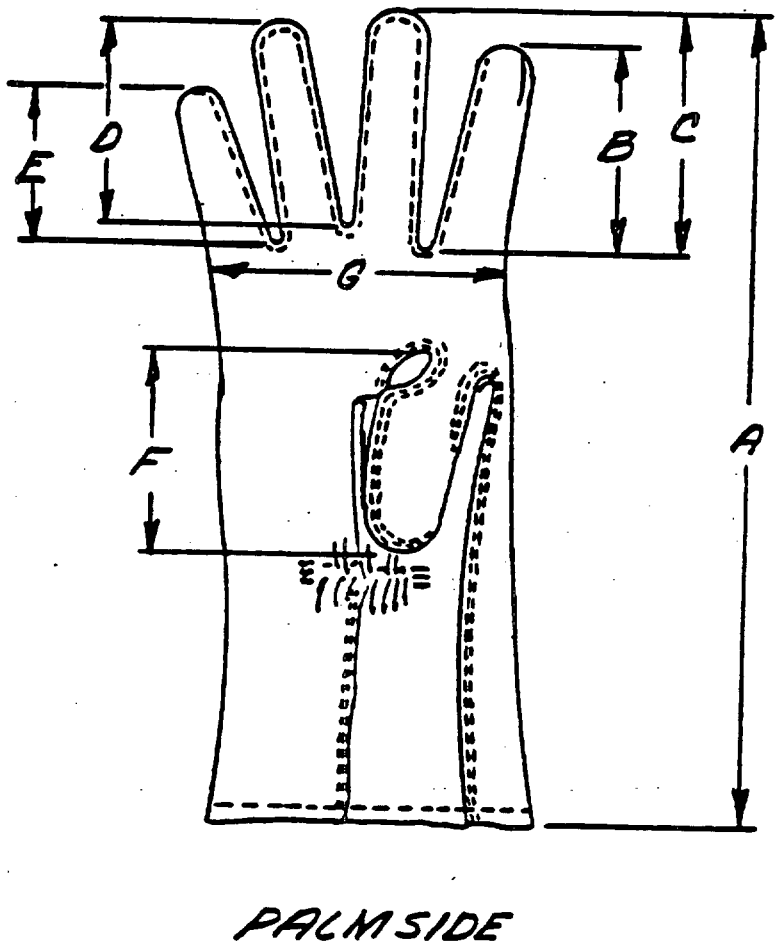



Figure 5. Gloves, Combat Vehicle Crewman's, Summer, Location of Dimensions.



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# STANDARD[illegible]ATION DOCUMENT IMPROVEMEN[illegible]ROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-G-44108A | Gloves, Combat Vehicle Crewman's, Summer |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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